Soil Health Management

Agriculture RAG Knowledge Base • Educational reference document

Soil health is the capacity of soil to function as a living system that supports plants, regulates water, cycles nutrients, and maintains biological activity. Good soil management focuses on physical structure, chemical fertility, organic matter, and biological diversity rather than treating fertilizer as the only input.

Key indicators include soil organic matter, pH, nutrient availability, aggregation, infiltration, water-holding capacity, earthworm activity, erosion, and compaction. Soil testing is useful because visual symptoms alone cannot reliably identify many nutrient deficiencies.

Practices that can improve soil health include adding suitable organic materials, maintaining crop residues, reducing unnecessary soil disturbance, using crop rotations, controlling erosion, and matching nutrient applications to crop requirements. Organic matter can improve aggregation and water retention, while diverse rotations can interrupt pest cycles and support nutrient cycling.

A practical soil-health workflow is: identify the crop and field history; collect representative soil samples; test pH and nutrients; identify limiting factors; choose management practices; apply inputs at appropriate rates and times; and monitor the response over subsequent seasons.

Soil problems differ by location. Acid soils may require liming where appropriate, saline soils require careful water and salt management, and compacted soils may benefit from reduced traffic and suitable tillage. Recommendations should be based on local soil tests and agronomic guidance rather than a single universal recipe.

Reference basis

• FAO, Integrated Soil Fertility Management

• FAO, Integrated Plant Nutrient Management

Note: This document is a project knowledge-base document prepared for a student RAG demonstration. It is not a substitute for crop-specific agronomic advice, soil testing, product labels, or local agricultural recommendations.